



COMSATS University Islamabad, Attock Campus
Department of Computer Science

Assignment 03

Deadline: 26th May 2025

Course: - Artificial Intelligence

Class: - BSE05

Total Marks: - 20

Semester: SP25

CLO-4: Apply resolution to a set of logic statements to answer a query

Question 01:

(10 marks)

You are developing the reasoning module for a household robot named AIRA (AI Robot Assistant). AIRA performs daily tasks like cooking, cleaning, alerting, and monitoring household safety based on environmental conditions and sensor input.

Some of the AIRA's actions are mentioned below. For each sentence:

- a. Convert it into **Propositional Logic (PL)** by clearly defining the atomic propositions.
- b. Convert these into **First Order Logic (FOL)** by defining predicates, functions, and constants, where applicable.

1. If AIRA detects smoke, it sounds an alarm.
2. If AIRA does not detect smoke, it does not sound an alarm.
3. AIRA is cooking if and only if the stove is on.
4. The stove is on.
5. Whenever AIRA is cooking, there is a risk of smoke.
6. If AIRA is cleaning, it moves around the house.
7. AIRA moves around the house only if there are no obstacles.
8. There is an obstacle in the living room.
9. AIRA cannot clean if there are obstacles in its path.
10. If the floor is wet, AIRA avoids the kitchen.
11. If AIRA avoids a room, it does not clean that room.
12. AIRA avoids any room that contains fire.
13. AIRA cleans the living room every day.
14. If the garbage bin is full, AIRA empties it.

15. AIRA only empties the bin when it is full.
16. If someone enters the house and AIRA is in guard mode, it alerts the user.
17. If the user is asleep, AIRA switches to silent mode.
18. If AIRA is in silent mode, it does not send alerts.
19. AIRA always recharges itself at night.
20. If AIRA is recharging, it cannot perform any tasks.
21. AIRA can only cook if it is not cleaning.
22. If the kitchen has gas leakage, AIRA disables the stove.
23. AIRA disables the stove if it detects excessive heat.
24. If AIRA disables the stove, it sends a notification.
25. AIRA never cleans the house when it is raining outside.
26. If any window is open during rainfall, AIRA sends a weather alert.
27. AIRA always locks the doors at night unless a family member is outside.
28. If AIRA detects motion at the door after midnight, it activates the security camera.
29. AIRA does not start vacuuming unless the noise level is below a certain threshold.
30. Every task AIRA performs must be approved by the scheduler module.

Question 02:

(10 marks)

A variety of intelligent agents (e.g., self-driving car, AI assistant, surveillance drone) operate based on conditional rules and sensor data. Given the knowledge base (KB), your task is to:

- Convert statements into **Propositional Logic (PL)**.
- Prove the specified conclusions using **either inference rules (like Modus Ponens, Modus Tollens, Hypothetical Syllogism) or resolution**.

(a) Represent all the given premises in PL and prove the conclusion using inference rules

Premises:

1. If the car is speeding, then the distance to the next vehicle is unsafe.
2. If the distance is unsafe, then the car should slow down.
3. The car is speeding.

Conclusion:

The car should slow down.

(b) Premises:

1. If the door is open, then the security alarm is triggered.
2. If the window is open, the door is open.
3. The window is open.

Conclusion:

The security alarm is triggered.

(c) Premises:

1. If motion is detected in a restricted area, then the drone captures an image.
2. No motion is detected.

Conclusion:

The drone does not capture an image.

(d) Premises:

1. If the battery is low, then the robot returns to the charging dock.
2. The robot did not return to the charging dock.

Conclusion:

The battery is not low.

(e) For the given statements first convert them into CNF and then proof by resolution.

Premises:

1. If a person is trapped, then the robot sends an alert.
2. If an alert is sent, the rescue team is notified.
3. A person is trapped.

Conclusion:

The rescue team is notified.